DAKSH SAWHNEY

QUANT · RESEARCHER · BUILDER

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EDUCATION

Inventure Academy | *Cambridge A Level* — Bangalore · 2024 – 2027

- **Grade 12 (in progress), predicted:** A* in Further Maths, Maths, Physics, Chemistry, Computer Science; A in English (AS)
- **Grade 11 (2026):** A* in Mathematics (A Level); A in Physics, Chemistry, Computer Science (AS)
- **IGCSE (2025):** 8 A* (Maths, Additional Maths, Physics, Chemistry, Biology, Economics, Computer Science, English Literature); A in English Language, Spanish; ICE Distinction
- **Courses:** Technical Analysis in Trading (Hexarum); Intro to Psychology (Yale); Intro to Data Science & AI (IIT Madras); UPenn FinTech Specialisation: Foundations, Payments & Regulations; Crypto & Blockchain; Lending, Crowdfunding & Modern Investing; AI, InsurTech & Real Estate Tech; Moneyball: AI, Sports, and Business (Inspirit AI)

RESEARCH

Research Science Institute – India | *Research Scholar, IISc (Dr. Kaushal Verma, Dean of Math)* — Bangalore · Jun – Jul 2026

- Selected as 1 of 30 students nationally (~2% acceptance) for MIT RSI-model programme; awarded 100% scholarship
- Studied transient chaos in nonlinear dynamical systems; engineered a bifurcation simulator
- Devised a structural coefficient (ξ) predicting bifurcation-diagram patterns from an equation's form; paper under review at JEI

Cambridge Centre for International Research | *Research Scholar, mentored under Dr. Weiwei Sun, Cambridge* — Nov 2025 – Mar 2026

- Tested an LLM part-of-speech tagger on Hinglish vs monolingual data, achieving 94.89% accuracy on code-mixed text
- Identified error cascades in mixed grammars, with most confusion among nouns, verbs, proper nouns, and adjectives
- Published paper in the Oxford Journal of Student Scholarship (OJSS), a peer-reviewed undergraduate journal (2026)

Independent Research | *UPI & Financial Inclusion in India* — Jul 2025

- Surveyed 140 respondents on UPI vs cash (64% found UPI more convenient); published in Int'l Journal of Social Science & Economic Research (2025)

PROJECTS & EXPERIENCE

Persifolio | *Founder & Lead Developer; Crest Gold Award Winner* — Bangalore · 2024 – Present

- Built a free AI investment simulator (Flutter, Firebase; live on Google Play) with personalised portfolios and risk scores
- Taught beginners and young adults investing, budgeting, and risk management, reaching 10,000+ users
- Secured a $50,000 pre-seed from Peacock Capital, alongside $25,000 of self-funded trading earnings; patent published in Govt of India IP Journal
- Endorsed by Govt of India; adopted by Odisha schools and as finance elective by partner St. Jerome's PU & Degree College

Independent Equity Trader | *Indian markets* — 2024 – Present

- Trade Indian equities using candlesticks, support/resistance, moving averages, RSI and MACD; self-invested $25,000
- Won 1st place at the HDFC Credelia Portfolio Strategy Challenge (2024); certified in Technical Analysis (Hexarum, 2024)

Polaris | *Founder & Lead Developer* — Jan 2026 – Present

- Co-built (team of 4) a free AI career coach learning from professionals in users' dream jobs; benchmarks CVs against real hires
- Enrolled 300+ students via 15+ workshops at 3+ partner orgs; built engine turning CVs + hiring data into 16-wk roadmaps; 1800+ gap tasks completed
- Won 1st Place at HackHarvard India; represented India at HackHarvard, an undergraduate hackathon at Harvard University, USA

ScamSlayer | *Creator, Developer & Workshop Leader* — Sep 2025 – Present

- Built a free gamified cybersecurity platform for seniors simulating fake-arrest and impersonation scams and phishing
- Trained 1,000+ residents across 20+ Bangalore elderly homes through workshops; averted ~$6K in fraud (est.)
- Featured in Outlook India (1.5M readership); lauded by a state lawmaker; patent published in Govt of India IP Journal

Nineleaps Technology Solutions Pvt. Ltd. | *AI Engineering Intern* — Bengaluru · Apr – May 2025

- Shipped a production multilingual AI-powered voice agent for live customer interaction; improved sales and reduced customer acquisition cost
- Built Flask/SQL services using language-model APIs; integrated LLM chatbots; explored RAG and vector embeddings for retrieval-based answers
- Performance rated "Higher than Expected" by the company; contributed to team deliverables and objectives under a supervisor

Macrospective | *Co-developer, HackHarvard, Harvard University (1 of 2 high-school teams selected)* — Oct 2026

- Built a Python multi-agent social media simulation of rumour-driven financial panics, driven by LLM-generated scenarios
- Tested which company responses limit the damage; forecast how announcements spread and affect sales

LEADERSHIP & ACTIVITIES

Shades of Tomorrow, Miyawaki Forest Project | *Lead Engineer and Youth Activist* — Bangalore · 2023 – Present

- Led 7 plantation drives across 6 Bangalore sites, planting 100,000+ saplings through the One Person One Tree campaign
- Raised INR 500,000 (~$6K) from corporate sponsors; engineered EcoWeave, an AI app mapping species synergies and growth

Competitive Athletics and Tech for Change Club | *Inventure Academy* — 2023 – 2026

- Captained varsity badminton team as Team Strategist; coached the squad on smash technique and doubles tactics with no coach
- Badminton: mixed adult doubles gold (Bangalore, 2023); State Open semi-finals; state U-17 singles semi-finals (2024); ISSO Nationals quarter-finals; won golds and silvers
- Shot put: interschool gold (2024), silver (2025), ISSO nationals; basketball: interschool silver (2024), gold; swim/relay: bronze
- Building an e-ballot platform for school elections as a Tech for Change member

Empower-Fin and RGH Government School | *Workshop Leader and Tutor* — 2024

- Taught inflation and money management to underserved students and women; tutored 5 students, lifting exam marks by 25%

SKILLS, HONORS, PUBLICATIONS & INTERESTS

Honors: 1st Place, HackHarvard India (Polaris, 2026); Rabindra Ratna Puraskar 2026 (Veyil Foundation, national top 15, financial literacy); #1 in grade for Computer Science; top-2 in grade for TechWhiz, Multi Achiever, Community Outreach; Crest Gold (2025); HDFC Credelia Strategy Challenge, 1st (2024)

Publications: "Sequence Labelling and Error Analysis for Code-Mixed Text," Oxford Journal of Student Scholarship (2026); "From Cash to Clicks: UPI and Financial Inclusion in India," Int'l Journal of Social Science & Economic Research (2025); Structural coefficient (ξ) for predicting bifurcation-diagram patterns in nonlinear dynamical systems, under review at JEI (2026)

Technical: Python, SQL, Flutter, Firebase, MongoDB, AWS, Azure, LLM APIs, multi-agent simulation, technical analysis

Interests: Stock trading, basketball, badminton, Himalayan hiking (Gomukh, 14,000 ft), Model UN